

PATH-SPACE COTANGENT RIGIDITY, UNIVERSAL CONTRACTIONS, AND TANGENT REPRESENTATIONS

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ABSTRACT. Invariant-integral and pressure null spaces are closed linear spans. A uniform Cesaro Lyapunov condition yields invariant Radon separators. Source-dependent Doob and memory response are transported by Kato parallel transport in one common transfer Banach bundle, avoiding nonexistent Radon–Nikodym derivatives between mutually singular equilibrium path measures.

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1. ROUND-EIGHT RECONSTRUCTION: CLOSED LINEAR NULL SPACES AND KATO TRANSFER BUNDLES

The invariant-integral null space is a closed linear span. Invariant separators are obtained only under a uniform Cesaro Lyapunov bound. Source response is transported in one common transfer-operator Banach bundle; no Radon–Nikodym derivative between mutually singular infinite-volume Gibbs measures is used.

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1.1. Two closed linear null spaces. Let $\mathcal{A}_W$ be the weighted path-observable Banach space. Define

$$\mathcal{N}_{\text{int}} = \overline{\text{span}}\{U - U \circ \Theta_t : U \in \mathcal{A}_W, t \geq 0\},$$

$$\mathcal{N}_{\text{pr}} = \mathcal{N}_{\text{int}} \oplus \mathbb{R}\mathbf{1}.$$

Theorem 1.1 (Invariant-integral quotient). *Suppose every separating weighted functional satisfies the uniform Cesaro Lyapunov estimate*

$$\sup_{T \geq 1} \frac{1}{T} \int_0^T \int W \circ \Theta_t d|\mu| dt < \infty.$$

Then $F - G \in \mathcal{N}_{\text{int}}$ exactly when all invariant probabilities have the same integral on F and G . Normalized pressure equivalence uses $\mathcal{N}_{\text{pr}}$ and therefore also quotients constants.

Proof. Every coboundary integrates to zero under an invariant law, and linear closure preserves this property. Conversely, separate $F - G$ from $\mathcal{N}_{\text{int}}$ by a weighted Radon functional μ . Average it:

$$\mu_T = T^{-1} \int_0^T (\Theta_t)_\# \mu dt.$$

The uniform Cesaro Lyapunov bound and compact sublevels of W give tightness. Any weak limit is invariant, annihilates every coboundary, and retains the nonzero pairing with $F - G$, contradicting equality of invariant integrals. Adding constants gives the normalized pressure quotient. $\square$

1.2. A common transfer-space bundle. Let $\mathcal{B}$ be one fixed strong/weak transfer scale after moving-chart identifications. The source-dependent Ruelle operators $\mathcal{L}_\eta : \mathcal{B}^+ \rightarrow \mathcal{B}^-$ form a C^m family with an isolated spectral projector P_η . Equilibrium measures are dual eigenvectors in $\mathcal{B}^*$; they need not be mutually absolutely continuous on infinite path space.

Lemma 1.2 (Kato parallel transport). *The equation*

$$U'_\eta = [P'_\eta, P_\eta]U_\eta, \quad U_0 = I,$$

has a bounded invertible solution on $\mathcal{B}$ and

$$P_\eta U_\eta = U_\eta P_0.$$

It transports the resolved spectral subspace without density square roots.

Proof. The commutator is a bounded continuous operator on the common scale, so the operator ODE has a unique product-integral solution. Differentiate $P_\eta U_\eta - U_\eta P_0$ and use $P_\eta^2 = P_\eta$; the derivative is the same homogeneous commutator equation with zero initial value, hence vanishes. $\square$

Define the covariant derivative of a resolved section by

$$\nabla_\eta s = P_\eta U_\eta \frac{d}{d\eta} (U_\eta^{-1} s).$$

Theorem 1.3 (Covariant Doob response without mutual absolute continuity). *The derivatives of the transfer eigenfunction, dual eigenfunctional, pressure, Doob operator, and resolved projector exist in the common Banach bundle and are obtained by the reduced resolvent and Kato connection. Finite-time likelihood cocycles implement the same derivative on path cylinders; no infinite-path Radon–Nikodym derivative is required.*

Proof. Differentiate the isolated Riesz projector and use the reduced resolvent on $(I - P_\eta)\mathcal{B}$. Normalize the right and left eigenvectors by their dual pairing. Kato transport identifies the moving ranges. On a finite path cylinder all source laws have positive densities relative to the same reference kernel, so ordinary likelihood differentiation is valid there. The uniform transfer estimates pass the finite-time formulas to the Banach-bundle limit. Distinct infinite-volume ergodic laws may remain mutually singular. $\square$

1.3. Causal memory and moving resolved subspaces. Let $C_\eta(z)$ be the compressed resolvent on $P_\eta\mathcal{B}$ and $A_\eta = P_\eta L_\eta P_\eta$. Define

$$\widehat{K}_\eta(z) = zP_\eta - A_\eta - C_\eta(z)^{-1}.$$

Theorem 1.4 (Covariant causal-memory derivative). *Under Kato transport,*

$$\nabla_\eta \widehat{K}_\eta = z\nabla_\eta P_\eta - \nabla_\eta A_\eta + C_\eta^{-1}(\nabla_\eta C_\eta)C_\eta^{-1}.$$

All connection terms from the moving projector and eigenvectors are included.

Proof. Differentiate the correctly signed identity $\widehat{K} = zP - A - C^{-1}$ in the common bundle. The inverse rule gives $\nabla(C^{-1}) = -C^{-1}(\nabla C)C^{-1}$, producing the plus sign in the last term. Lemma 1.2 supplies the derivative of the moving range, and Theorem 1.3 supplies the eigenvector and pairing terms. $\square$

1.4. Optional projections. For the A4 past filtration let

$$M_t^\varepsilon = \frac{\mathbb{E}[\exp\{\theta\Phi(X_T^\varepsilon)\} \mid \mathcal{F}_t^-]}{\mathbb{E}\exp\{\theta\Phi(X_T^\varepsilon)\}}.$$

Theorem 1.5 (Stable optional-projection convergence). *The exact likelihood martingales converge jointly with the A4 past kernels to the diffusion optional projection, stably under bounded initial-past multipliers.*

Proof. Write the conditional expectation as the A4 past-kernel integral of the terminal functional. Uniform quenched kernel convergence on compact past sets gives convergence there; exponential containment removes the complement. Multiplication by a bounded initial-past variable is absorbed into the same kernel integral, which proves stable convergence. $\square$

1.5. Nonlinear contractions without a fictitious adjoint. For a continuous nonlinear contraction $\mathcal{C} : X \rightarrow Y$, define the test pullback

$$\mathcal{C}^\sharp \phi = \phi \circ \mathcal{C}.$$

Theorem 1.6 (Typed pressure–rate contraction chain rule). *If Q_X is differentiable on a source chart, then*

$$Q_Y(\phi) = Q_X(\mathcal{C}^\sharp \phi), \quad DQ_Y(\phi)h = DQ_X(\phi \circ \mathcal{C})(h \circ \mathcal{C}).$$

The rate on Y is

$$I_Y(y) = \inf_{\mathcal{C}(x)=y} I_X(x).$$

For smooth parameterized contractions, the next derivative uses the ordinary Frechet chain rule for $\phi \circ \mathcal{C}_\eta$.

Proof. The Laplace functional of the pushforward is the Laplace functional of the pulled-back test. Differentiate this identity on the declared source chart. The rate formula is the contraction principle. No linear adjoint of the nonlinear map $\mathcal{C}$ is introduced. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared local source, short-time, finite-horizon or Boltzmann–Grad regime specified in its theorems. Every load-bearing cross-paper input is typed in the round-eight dependency ledger. Clean compilation and automated structural checks are necessary verification gates; independent external journal review remains a separate process.

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